

Assignment # 3

Date: _____

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erp id: 32442

Q1.

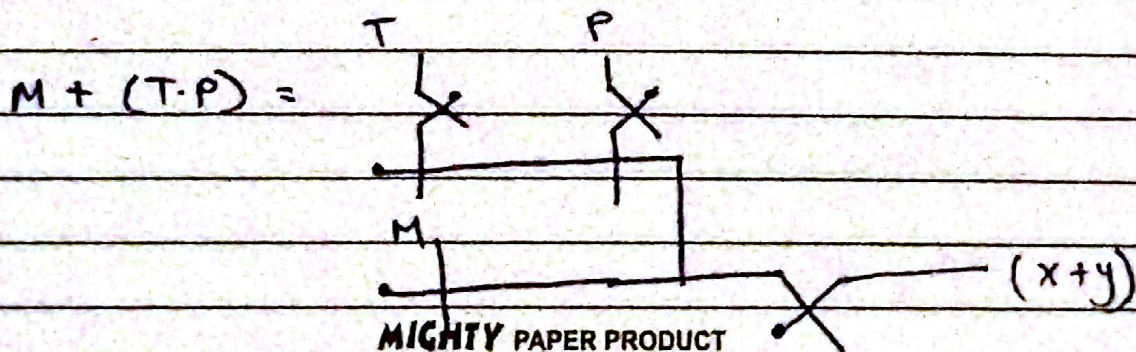
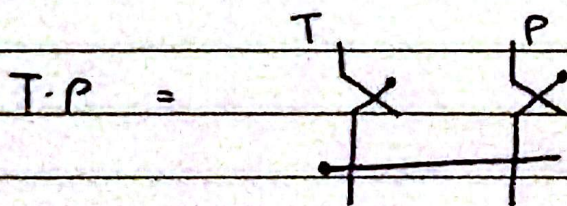
light will turn on when the master switch of the circuit is on OR the temperature $> 30^{\circ}\text{C}$ AND pressure > 50

truth table

M	T	P	light
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$\text{light} = (M) \text{ OR } (T \text{ AND } P)$$

$$M + (T \cdot P)$$



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Q2-

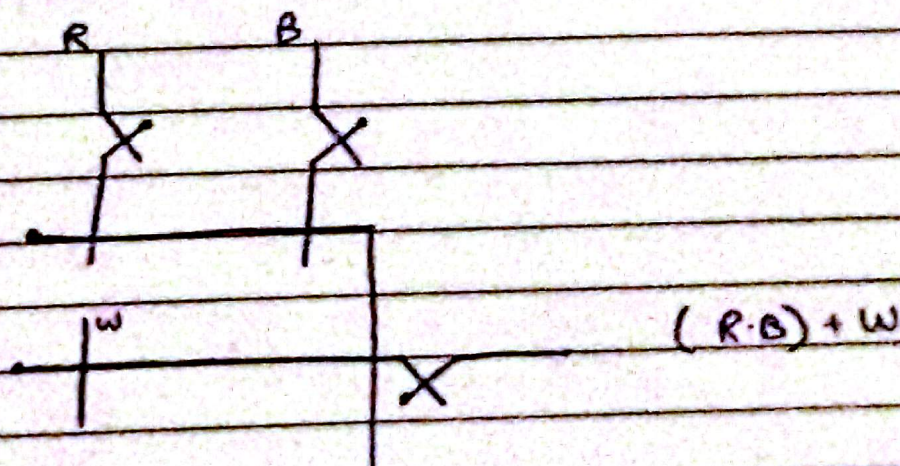
Pump will output 1 if the water level is greater than 10cm OR if rain is expected AND backup battery is fully charged

W	R	B	activation
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Pump activation :

$$(W) \text{ OR } (R \text{ AND } B)$$

$$W + (R \cdot B)$$



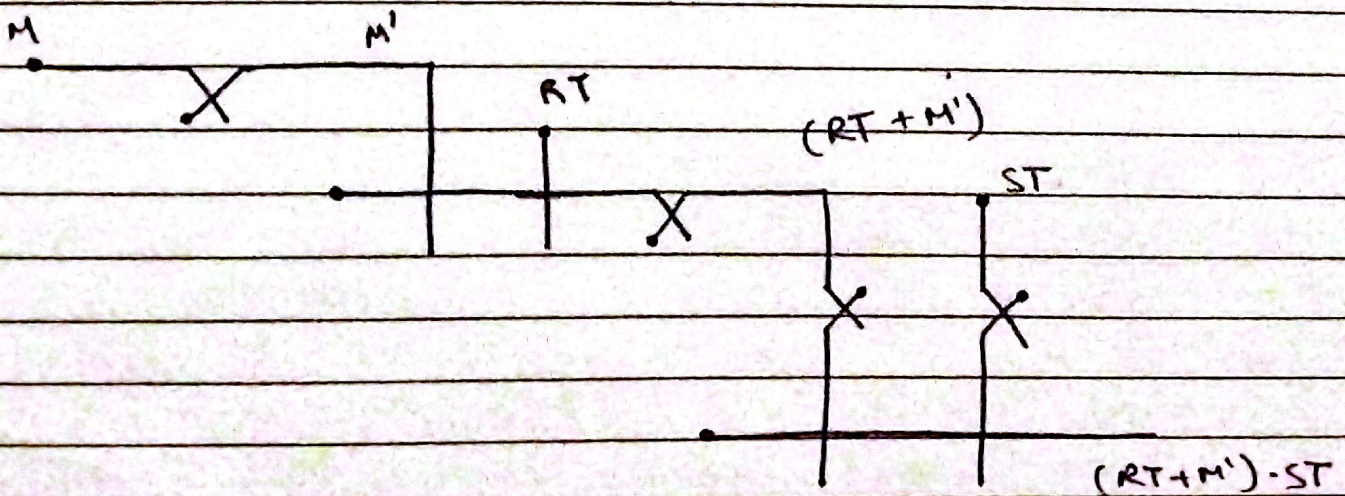
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Q3- fan turns on when temp $> 80^{\circ}\text{C}$ AND maintenance switch is NOT active OR if temp $> 80^{\circ}\text{C}$ AND room temperature $> 25^{\circ}\text{C}$

ST	RT	MS	fan
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Boolean :- (ST AND (NOT M)) OR (RT AND ST)
 expression $(ST \cdot M') + (RT + ST)$
 $ST \cdot (RT + M')$



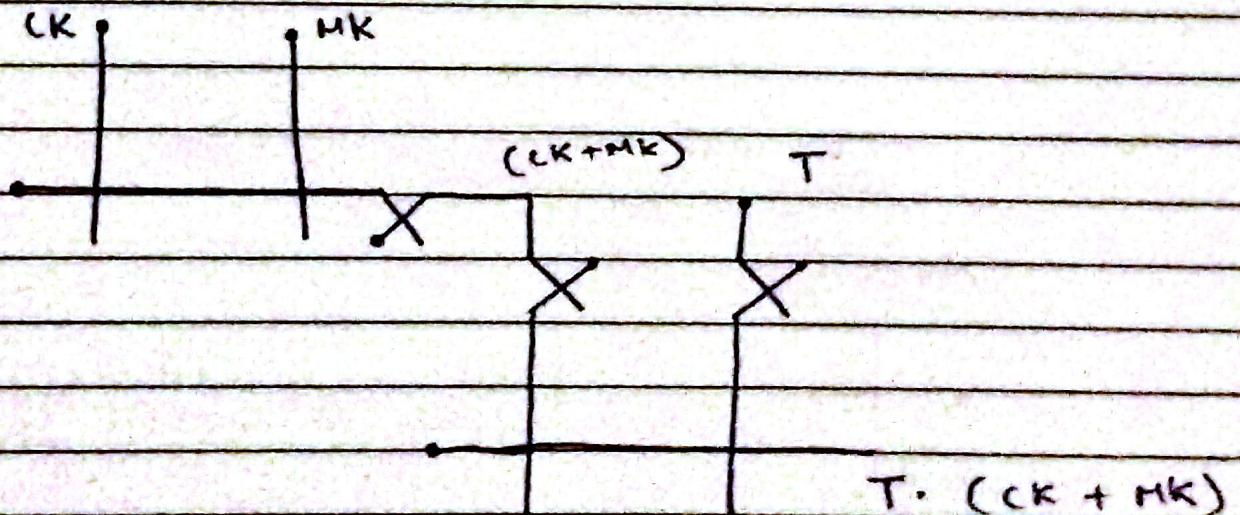
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Q4- Vault lock outputs a 1 if time lock is activated AND atleast 1 of cashiers key OR managers key is inserted

T	CK	MK	lock
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression:

$$T \cdot (CK + MK)$$



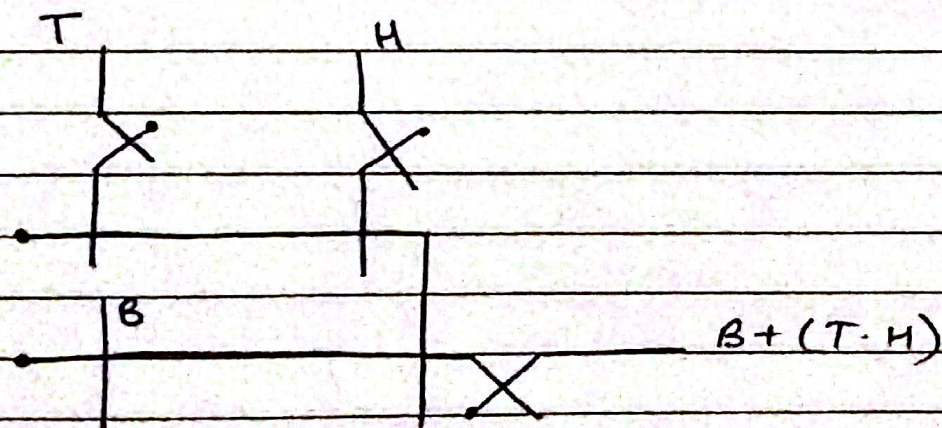
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Q5- defroster will activate if button is pressed manually
OR if temp is $< 40^{\circ}\text{C}$ AND humidity is $> 70\%$.

B	T	H	activation
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression:

$$(B) \text{ OR } (T \text{ AND } H)$$
$$B + (T \cdot H)$$



Date: _____

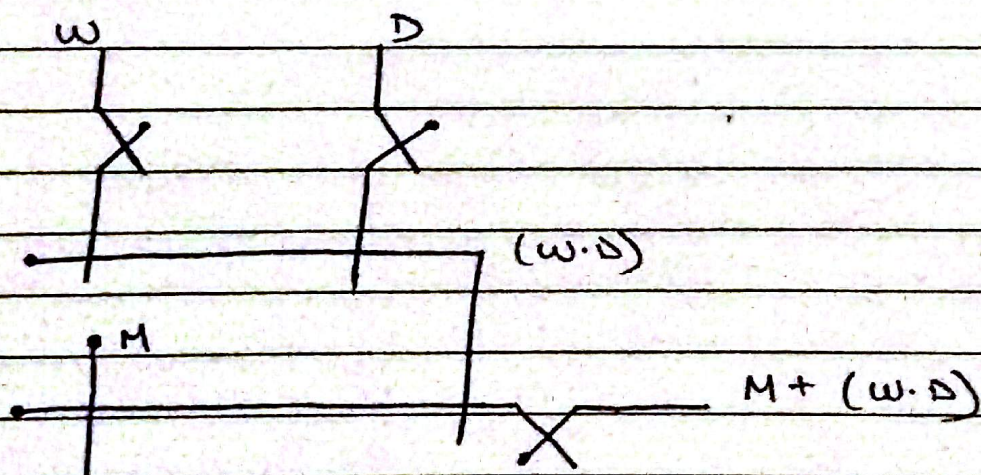
Q16- package is rejected if a foreign metal is detected and outputs a 1 OR if the package is > 5kg AND too full

weight	D	Metal	output
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression:

$$(M) \text{ OR } (W \text{ AND } D)$$

$$M + (W \cdot D)$$



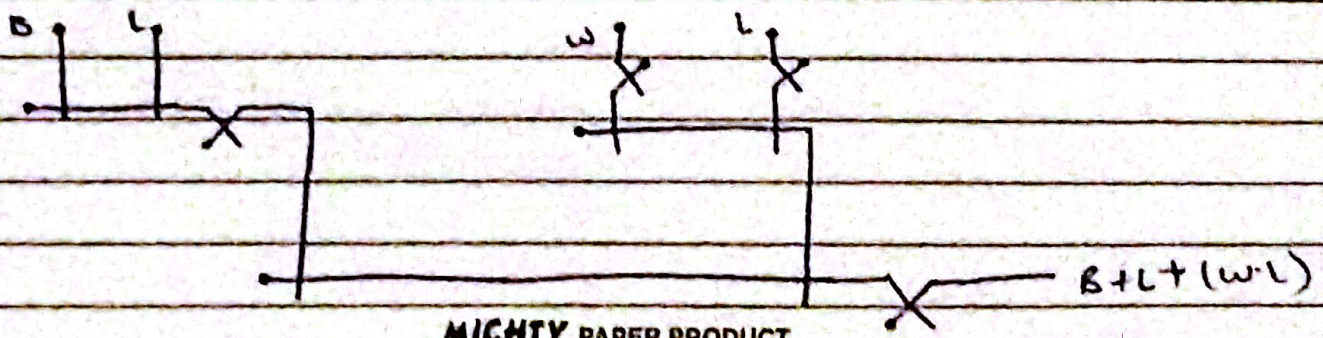
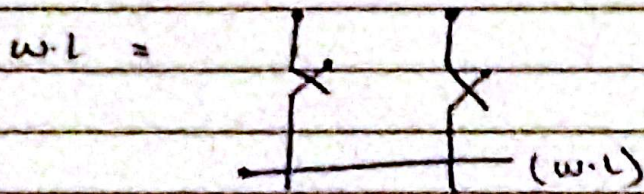
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Q7. Reopens if bumper strikes an object OR if lower tripwire is broken OR reopens if weight > 1000 kg AND tripwire is broken

L	W	TB	door
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression:

$$B + L + (W \cdot L)$$



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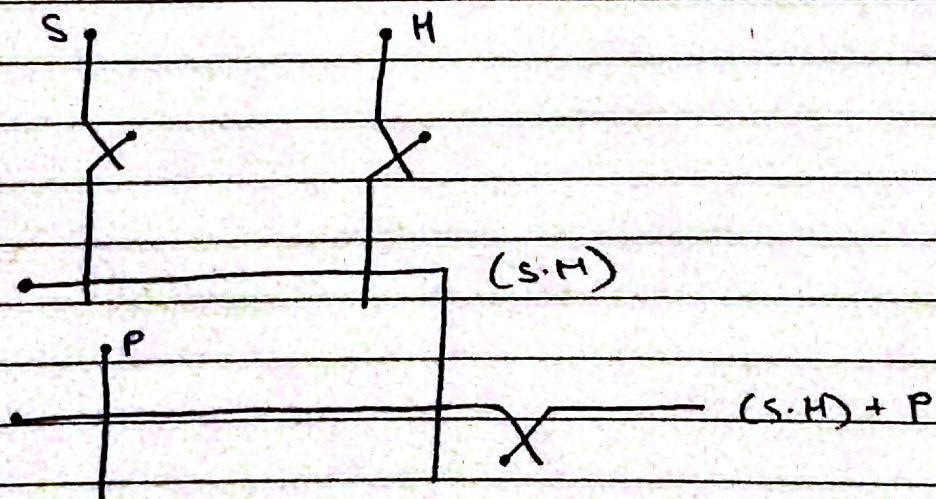
Q8- detector gets triggered by manual pull station OR if heat is 77.5°C AND smoke opacity $> 20\%$.

S	H	P	Sprinkler
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression

$$(S \text{ AND } H) \text{ OR } (P)$$

$$(S \cdot H) + P$$



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Q9- output is 1 if its night time AND all chickens are inside OR its daytime AND raining AND chickens inside

L = daytime (1 if light > 500 lux)

M = all chickens inside (1)

R = raining (1)

D = door closes (1)

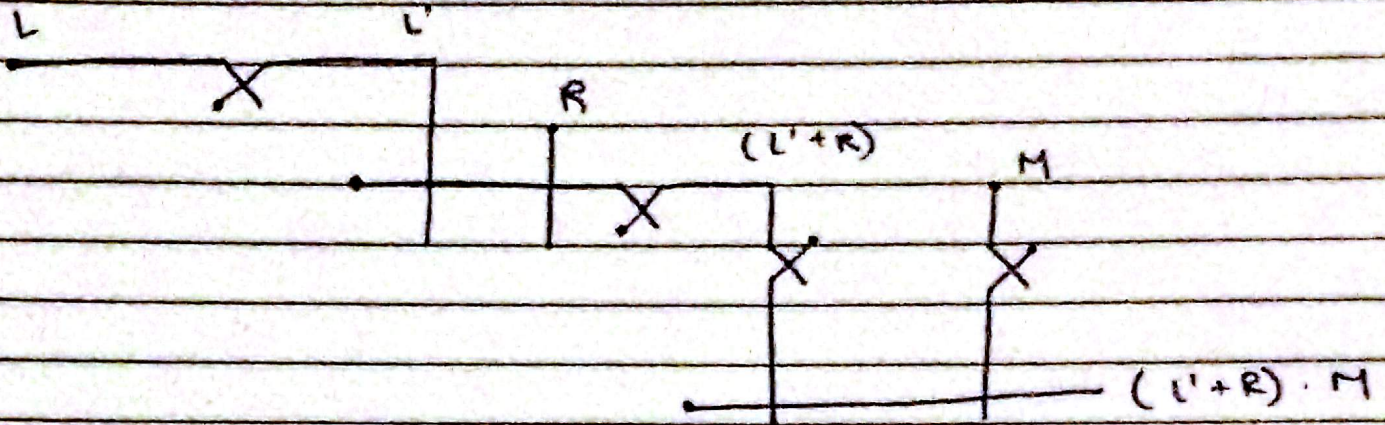
L	M	R	D
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$D = (M \text{ AND } (\text{NOT } L)) \text{ OR } (M \text{ AND } L \text{ AND } R) \\ (M \cdot L') + (M \cdot L \cdot R)$$

simplifies to

$$D = M \cdot (L' + R)$$

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Q10- Activates if battery $< 15\%$ AND distance $> 2\text{km}$ OR
if battery $< 15\%$ AND wind $> 20\text{km/h}$

B = battery $< 15\%$ (1)

D = distance $> 2\text{km}$ (1)

W = Wind $> 20\text{km/h}$ (1)

S = Saver Mode activates (1)

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B	D	w	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression =

$$S = (B \text{ AND } D) \text{ OR } (B \text{ AND } w)$$

$$S = (B \cdot D) + (B \cdot w)$$

$$S = B \cdot (D + w)$$

